HOSPITAL ANALYTICS DASHBOARD

An End-to-End Data Analytics Project

PREPARED BY

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TOOLS & TECHNOLOGIES

Python | Pandas | NumPy | MySQL | Microsoft Excel 365 | Microsoft Power BI

ACADEMIC YEAR

2025 - 2026

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1. Introduction

Healthcare organizations generate large volumes of patient data every day, including admission details, department information, waiting times, billing records, emergency cases, and patient satisfaction levels. Analyzing this data manually is time-consuming and makes it difficult to identify operational issues and improve patient services.

The Hospital Analytics Dashboard project demonstrates an end-to-end data analytics workflow using Python, SQL, Microsoft Excel, and Microsoft Power BI. The project converts raw hospital data into meaningful insights through data cleaning, SQL-based analysis, KPI generation, and interactive dashboards.

These insights help hospital management improve operational efficiency, monitor patient care, and support data-driven decision-making.

2. Business Problem

Hospitals handle thousands of patient records every month. Without proper analytics, it becomes difficult to monitor hospital performance and identify operational improvements. Some common challenges include:

- Difficulty in analyzing patient waiting times.
- Lack of visibility into departmental performance.
- Challenges in monitoring hospital billing trends.
- Limited understanding of patient satisfaction levels.
- Manual reporting consumes significant time and effort.

To solve these challenges, this project develops a complete hospital analytics solution using Python, SQL, Excel, and Power BI.

3. Project Objectives

The objectives of this project are:

- Clean and preprocess raw hospital data.
- Improve data quality by handling missing and inconsistent values.
- Generate meaningful healthcare KPIs.
- Perform SQL-based business analysis.
- Develop Excel dashboards for reporting.
- Create interactive Power BI dashboards.
- Analyze hospital operations and patient trends.
- Support data-driven healthcare management.

4. Dataset Description

The dataset used in this project was collected from Kaggle and contains hospital patient records.

Dataset Details

Attribute	Description
Dataset Source	Kaggle
Dataset Type	CSV
Domain	Healthcare Analytics
Total Records	Approximately 10,000

Important Columns

- Patient ID
- Department
- Waiting Time (Minutes)
- Total Bill
- Emergency Case
- Satisfaction Level
- Age
- Gender
- Diagnosis
- Admission Date

The dataset was cleaned and transformed before performing business analysis.

5. Data Cleaning using Python

Python was used to clean and preprocess the hospital dataset.

Libraries Used

- Pandas
- NumPy
- Matplotlib

Data Cleaning Steps

- Imported CSV dataset.
- Identified missing values.
- Handled null values in important columns.
- Removed duplicate records.
- Standardized data types.
- Cleaned department names.
- Verified waiting time values.
- Validated billing information.
- Exported cleaned dataset as clean_hospital.csv.

The cleaned dataset was then used for SQL analysis, Excel reporting, and Power BI dashboard development.

6. SQL Business Analysis

Business analysis was performed using MySQL.

SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- LIMIT
- Aggregate Functions
- String Functions
- Date Functions

Business Queries Performed

- Average Waiting Time by Department
- Total Hospital Bill by Department
- Patients by Satisfaction Level
- Average Bill by Emergency Case
- Department Performance Analysis

These SQL queries transformed raw hospital records into meaningful healthcare insights.

7. Excel Dashboard

Microsoft Excel 365 was used to build an interactive dashboard using Pivot Tables, Pivot Charts, and KPI Cards.

Dashboard Components

- Total Patients
- Average Waiting Time
- Average Hospital Bill
- Highest Bill
- Lowest Bill
- Emergency Patients
- Average Bill by Department
- Average Waiting Time by Department
- Patients by Satisfaction Level
- Emergency Case Analysis

The dashboard provides a quick overview of hospital performance and patient statistics.

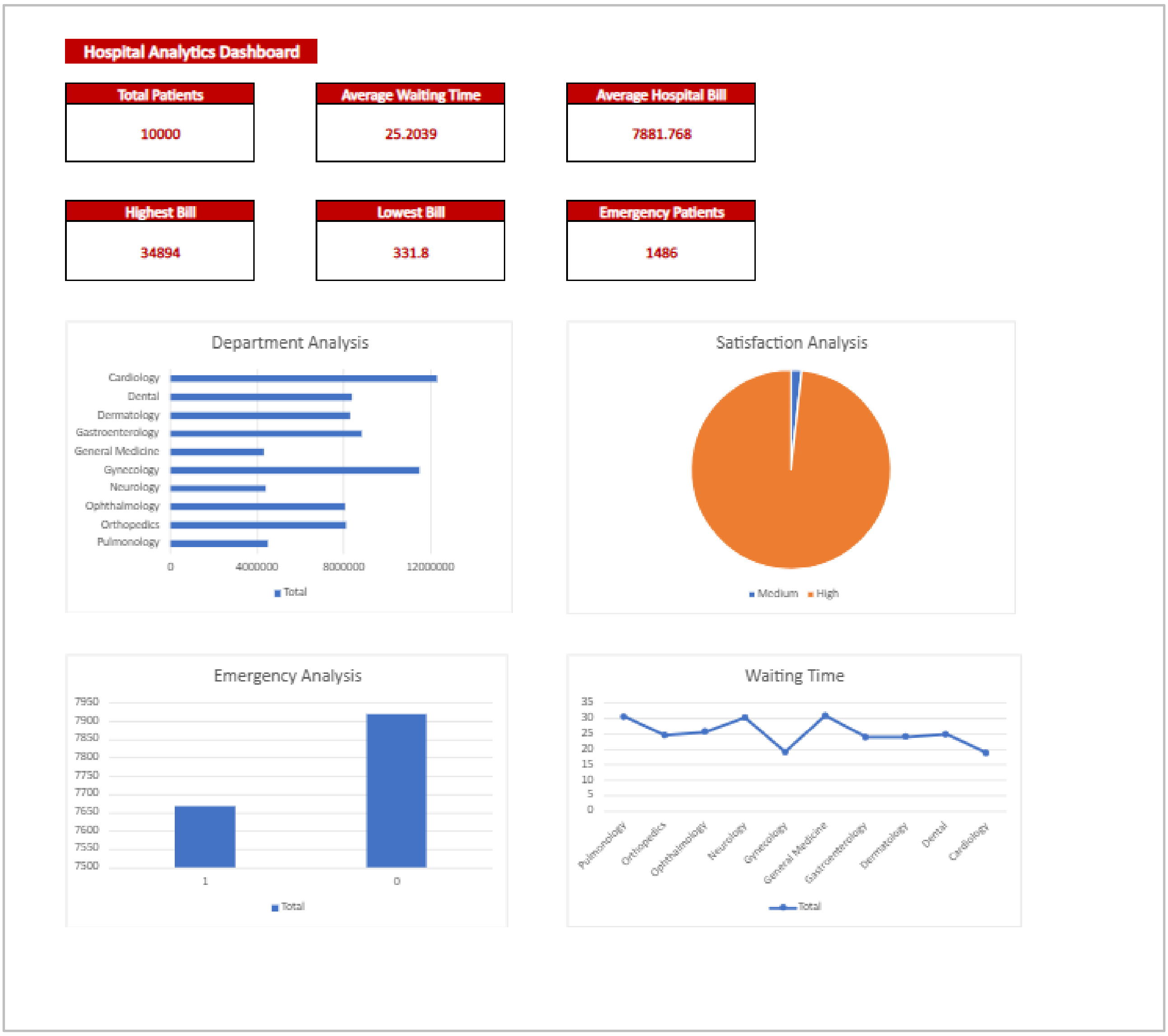

Figure 1: Excel Hospital Analytics Dashboard

8. Power BI Dashboard

Microsoft Power BI was used to create an interactive healthcare dashboard.

Dashboard Components

- KPI Cards
- Average Hospital Bill by Department
- Average Waiting Time by Department
- Patients by Satisfaction Level
- Emergency Cases

The Power BI dashboard enables hospital administrators to monitor operational performance through interactive visualizations and business intelligence.

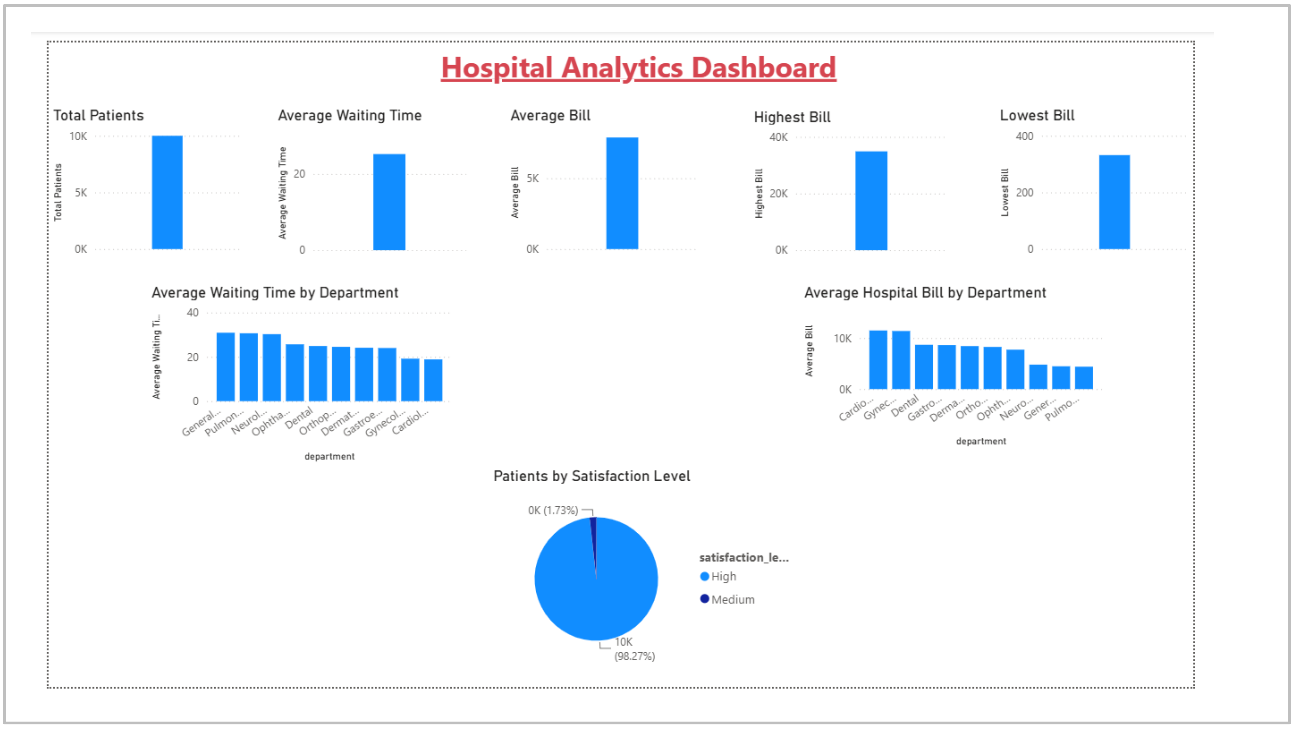

Figure 2: Power BI Hospital Analytics Dashboard

9. Key Performance Indicators (KPIs)

The following KPIs were generated during the project:

KPI	Value
Total Patients	10,000
Average Waiting Time	25.20 minutes
Average Hospital Bill	7,881.77
Highest Bill	34,894
Lowest Bill	331.80
Emergency Patients	1,486

These KPIs summarize the hospital's operational performance and patient service metrics.

10. Business Insights

The analysis generated several valuable healthcare insights:

- Some departments experienced higher patient waiting times than others.
- Hospital billing varied across different departments.
- Emergency cases represented a significant portion of patient admissions.
- Patient satisfaction levels provided valuable feedback on healthcare services.
- Dashboard analytics supported better hospital resource planning and operational improvements.

These insights help hospital management improve efficiency and patient experience.

11. Technologies Used

Technology	Purpose
Python	Data Cleaning & Preprocessing
MySQL	Business Analysis
Microsoft Excel	Dashboard Development
Microsoft Power BI	Interactive Dashboard
GitHub	Version Control

12. Folder Structure

```
Hospital_Analytics/     ■■■ Dataset/     ■■■ Documentation/     ■■■
Excel/     ■■■ Images/     ■■■ PowerBI/     ■■■ Python/     ■■■ SQL/
    ■■■ README.md
```

13. Learning Outcomes

This project helped in developing the following skills:

- Healthcare data preprocessing using Python.
- SQL-based analytical query writing.
- KPI generation for hospital datasets.
- Dashboard development using Microsoft Excel.
- Interactive visualization using Microsoft Power BI.
- Healthcare data interpretation.
- End-to-end data analytics workflow.

14. Future Scope

The project can be enhanced by:

- Integrating real-time hospital databases.
- Developing predictive models for patient admission forecasting.
- Predicting patient waiting times using Machine Learning.
- Publishing dashboards using Power BI Service.
- Building AI-powered hospital performance monitoring systems.

15. Conclusion

The Hospital Analytics Dashboard project successfully transformed raw healthcare data into meaningful business insights using Python, SQL, Microsoft Excel, and Microsoft Power BI. The project demonstrates a complete data analytics workflow, including data preprocessing, business analysis, dashboard development, and visualization.

The generated dashboards help healthcare organizations monitor operational performance, improve patient care, optimize resource allocation, and support data-driven decision-making.

16. References

- Kaggle Dataset Repository
- Python Documentation
- Pandas Documentation
- MySQL Documentation
- Microsoft Excel Documentation
- Microsoft Power BI Documentation